Resolución de Problemas
Parte III

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6 de mayo de 2025

Contenido

Ejercicios de https://onlinejudge.org/

- UVA 10074
- UVA 437

UVA 10074

The poor man went to the King and said, "Lord, I cannot maintain my family. Please give me some wealth so that I can survive with my wife and children." The King replied, "I shall grant you a piece of land so that you can cultivate and grow food for your family. In the southern part of the Kingdom there is a rectangular forest. Trees have been planted there at regular intervals. Some of the trees have been cut for use. You are allowed to take any rectangular piece of land that does not contain any tree. You need not go to the forest to select the piece of land. I have a map containing 1's at places where there is a tree and 0's at points where the tree has been cut." Help the poor man to nd out the largest piece of land. Area of the land is measured in units of number of trees that were there. Your program should take a matrix of 1's and 0's as input and output the area of the largest rectangular piece of land that contain no tree. Be careful about the efficiency of your program.

67
0110110
0000010
1000001
0100001
1100010
1101100

UVA 10074

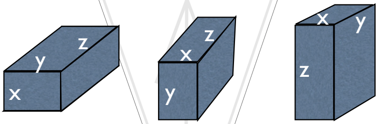
Este es un problema de mínima suma o en realidad de suma 0, y verificar cual es la matriz mas grande de suma cero.
Similar a maxsum2D

UVA 437

Perhaps you have heard of the legend of the Tower of Babylon. Nowadays many details of this tale have been forgotten. So now, in line with the educational nature of this contest, we will tell you the whole story: The babylonians had n types of blocks, and an unlimited supply of blocks of each type. Each type-i block was a rectangular solid with linear dimensions (x_i, y_i, z_i). A block could be reoriented so that any two of its three dimensions determined the dimensions of the base and the other dimension was the height. They wanted to construct the tallest tower possible by stacking blocks. The problem was that, in building a tower, one block could only be placed on top of another block as long as the two base dimensions of the upper block were both strictly smaller than the corresponding base dimensions of the lower block. This meant, for example, that blocks oriented to have equal-sized bases couldn't be stacked. Your job is to write a program that determines the height of the tallest tower the babylonians can build with a given set of blocks.

UVA 437

en este caso se tienen distintos tipos de bloques de diferente tamaño (x,y,z), que pueden ser usados varias veces y pueden ser usados de distinta forma:



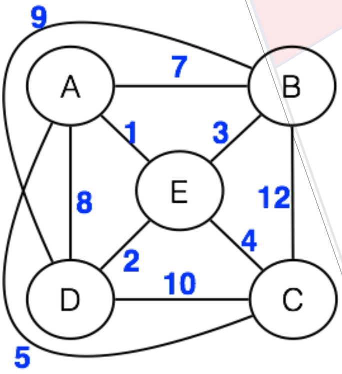
TSP

El problema del vendedor viajero, consiste en encontrar el camino de menor distancia; que permita visitar todas las ciudades de una zona, partiendo de una ciudad particular y terminando en la misma. La única ciudad que se puede repetir es la ciudad origen.

Se cuenta con la información de las ciudades y la información de las distancias entre cada par de ciudades.

Ejemplo TSP

Un ejemplo de solución al problema por búsqueda exhaustiva seria:



{A,B,C,D,E,A}=32	{A,D,B,C,E,A}=34
{A,B,C,E,D,A}=33	{A,D,B,E,C,A}=29
{A,B,D,C,E,A}=31	{A,D,C,B,E,A}=34
{A,B,D,E,C,A}=27	{A,D,C,E,B,A}=32
{A,B,E,C,D,A}=32	{A,D,E,B,C,A}=30
{A,B,E,D,C,A}=27	{A,D,E,C,B,A}=33
{A,C,B,D,E,A}=29	{A,E,B,C,D,A}=34
{A,C,B,E,D,A}=30	{A,E,B,D,C,A}=28
{A,C,D,B,E,A}=28	{A,E,C,B,D,A}=34
{A,C,D,E,B,A}=28	{A,E,C,D,B,A}=31
{A,C,E,B,D,A}=29	{A,E,D,B,C,A}=29
{A,C,E,D,B,A}=27	{A,E,D,C,B,A}=32

Multiplicación de matrices

El problema de multiplicar una serie de matrices es uno complejo, ya que depende de la cantidad de productos escalares que se deben realizar para obtener la matriz resultante.
La operación de la multiplicación, es una operación binaria.
El proceso se hace por agrupación

Ejemplo Multiplicación de 3 matrices

Si se quiere multiplicar tres matrices:

$$A_1 * A_2 * A_3 = \begin{cases} (A_1 * (A_2 * A_3)) \sim 300000 \\ ((A_1 * A_2) * A_3) \sim 15000 \end{cases}$$

$$A_1 = M_{10*200}$$
$$A_2 = M_{200*5}$$
$$A_3 = M_{5*100}$$

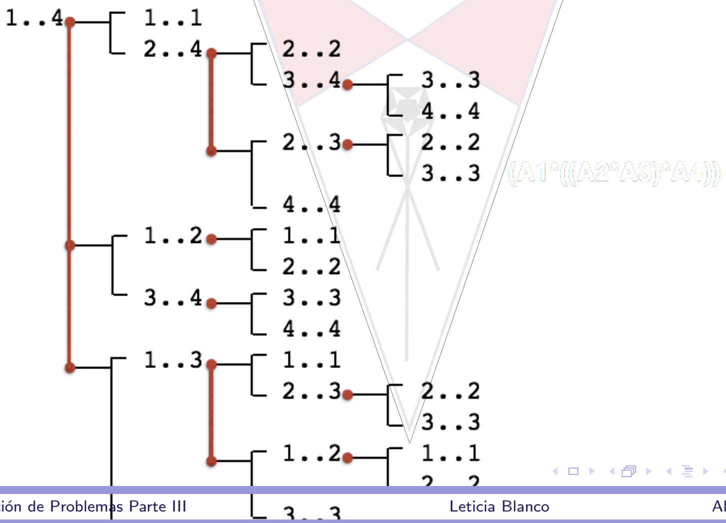
Multiplicación de matrices

Debido a que la multiplicación es una operación binaria, la cantidad de posibles distintas formas de agrupar esta dada por la siguiente relación:

$$n = \text{número de matrices}$$

$$nroAgrup = \sum_{k=1}^{n-1} (P(k)*P(n-k))$$

Agrupaciones para Multiplicar 4 matrices



Costo para Multiplicar 4 matrices